

Sushil Bhattarai

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EDUCATION

University of Southern Mississippi - Sophomore

Bachelor of Science in Computer Science, GPA: 4.00

Hattiesburg, MS

Aug. 2025 – Present

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL (MySQL), PHP

Frameworks / Libraries: React, Next.js, Node.js, React Native, Expo

Tools / Technologies: Git, Docker, Nginx, Kafka, Redis, BullMQ, Firebase, MongoDB

Deployment: Amazon AWS EC2, Digital Ocean Droplets, Linode VPS

Strong Interest: Scalable Distributed System Development

FEATURED PROJECT

Ehulak.tech — Distributed Email Delivery Platform

AWS EC2, Node.js, Kafka, Redis, BullMQ, Docker, Nginx, MongoDB, SMTP [Live Link]

- ★ Built an API-first email delivery platform that enables applications to send and track emails through a reliable queue → worker → inbox processing pipeline.
- ★ Designed an event-driven architecture using Redis (BullMQ) and Kafka to decouple services and support scalable asynchronous message processing.
- ★ Implemented worker based SMTP connection pooling, rate limiting, concurrency controls, retry mechanisms, and Dead Letter Queue (DLQ) recovery to improve reliability and fault tolerance.
- ★ Containerized services using Docker and deployed behind Nginx on a VPS, providing real-time tracking across the entire email lifecycle from queued to delivered states.

EXPERIENCE

Software Engineer Intern

Sep. 2023 – Jun. 2024

Z1 Technologies (Z1 GPS) [Live App]

- Developed a React Native mobile application using real-time data streaming via the Traccar API for continuous vehicle location updates.
- Implemented live map visualizations for streamed GPS coordinates, speed, distance, and route data.
- Integrated Firebase Cloud Messaging (FCM) for push notifications and remote vehicle control features in a production system integrated over 1,000 vehicles.

PROJECTS

PassMyFiles.com — Quick File Sharing Platform

Node.js, Kafka, Redis, BullMQ, AWS EC2, AWS S3, Docker, Nginx [Live Link]

- ★ Built a secure file-sharing platform enabling users to upload, store, and generate shareable links for large files.
- ★ Designed a distributed backend architecture using Kafka and BullMQ workers to handle asynchronous file processing and background tasks.
- ★ Implemented AWS S3 for scalable object storage and EC2 deployment behind Nginx for production-grade hosting and routing.
- ★ Integrated Redis for caching file metadata and link management, improving retrieval speed and system efficiency.

Ragat Nepal

React Native, PHP, MySQL, Expo [Live App]

- ★ Built a real-time blood donor matching platform for emergency response scenarios used by over 3,000 users with 1,000+ Google Play downloads.
- ★ Implemented location-based matching and notification workflows to support faster donor coordination.